

SOFTWARE REQUIREMENTS SPECIFICATION

Student Record Management System (SRMS)

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1. Introduction

1.1 Purpose

This document specifies the requirements for the Student Record Management System (SRMS). It serves as a formal reference for developers, instructors, and evaluators.

1.2 Scope

SRMS is a web-based application that manages student academic records with authentication, role-based access, reporting, and data analytics using a C shared library for optimized processing.

2. Overall Description

The system is implemented using Flask (Python), SQLite DBMS, and a C-based shared library integrated via ctypes. The application supports Admin, Teacher, and Student roles.

3. Functional Requirements

- User authentication and authorization
- Student record CRUD operations
- Sorting, searching, and statistical analysis using C DSA library
- PDF and Excel export functionality
- Graphical visualization of student marks

4. Non-Functional Requirements

- Performance: Optimized data processing using C
- Security: Hashed password storage and session control
- Portability: Cross-platform web application
- Maintainability: Modular architecture

5. External Interfaces

- Web UI (HTML/CSS)
- Database Interface (SQLite)
- C Shared Library Interface (ctypes)

6. Future Enhancements

- Subject-wise records
- Cloud database support
- Mobile responsiveness

- REST API integration

7. Conclusion

The SRMS demonstrates effective integration of OS concepts, DBMS, DSA in C, and modern web technologies to deliver a scalable academic management solution.